MIL-C-19087A(OS)
27 August 1970
Superseding
MIL-C-19807(NOrd)
16 September 1955

MILITARY SPECIFICATION

CONTRACTORS

(For Torpedo Mark 37 Mods 0, 2, and 3)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification describes the requirements for the manufacture, procurement, inspection, and test of series and paralleling contactors, NAVORD Drawings 877137 and 877138.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal.

PPP-B-621	Boxes, Wood, Nailed and Lock Corner

Military

MIL-P-116	Preservation, Methods of
MIL-V-173	Varnish, Moisture- and Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment)
MIL-R-2033	Relays for Naval Shipboard Electrical Service
MIL-M-18848	Motor, Drive, Torpedo, Counter-Rotating

FSC 1355

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:16Z
Check the source to verify that this is the current version before use.

MIL-B-18855	Battery Mark 53 Mod 0, Secondary Type, Silver-Zinc-Alkali
MIL-I-24092	Insulating Varnish, Electrical, Impregnating
MIL-E-82590	Environmental Requirements and Test 6 (For Torpedo Mk 37 Components and Assemblies)

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet Type Base

DRAWINGS

Naval Ordnance Systems Command

NAVORD LD 284040	Paralleling Contactor Assembled, and all parts and subassemblies listed thereon
NAVORD LD 284041	Series Contactor Assembled, and all parts and subassemblies listed thereon
NAVORD Drawing 815178	Coil Subassembly
NAVORD Drawing 865918	Hinge Subassembly
NAVORD Drawing 865924	Washer
NAVORD Drawing 865968	Counter Balanced Armature
NAVORD Drawing 877132	Battery Compartment Assembled
NAVORD Drawing 877137	Series Contactor Assembled
NAVORD Drawing 877138	Paralleling Contactor Assembled
NAVORD Drawing 1154934	Counter Balanced Armature

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:16Z
Check the source to verify that this is the current version before use.

PUBLICATIONS

Naval (Ordnance Systems Command

NAVORD OS 6341	Ordnance Design Requirements, General, Miscellaneous

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS.

Handbook H28	Screw Thread Standards for Federal Services

(Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Description. The two contractors described herein are electrically operated switching devices for intermittent duty, across the line, torpedo motor starting service and are mechanically and physically alike except for their contact and interlock configuration. The series contactor is a single-pole, single-throw, double-break type, while the paralleling contactor has a double-pole, single-throw, single-break arrangement. The armatures of both contractors have knife edge type hinges and are counterbalanced and mechanically interlocked by a linkage which effectively prevents both contractors from being closed simultaneously. The contractors are designed for mounting with their bases lying in vertical, parallel planes, and their contacts uppermost as shown on NAVORD Drawing 877132. The contactor coils are designed for operation at 24 volts direct current (dc), nominal. The contact feces are so designed as to limit the arcing to the "toe" of the contacts while the "heel" of the contacts carry the current. The contacts are fabricated from sintered silver and cadmium oxide powders and are so mounted that the moving contact will rock during closing and opening so as to minimize welding and sticking. An are shield is provided to assist in extinguishing the are during current interruption.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:16Z
Check the source to verify that this is the current version before use.

3.2 General requirements

3.2.1 Data required. No data is required by this specification (other than reports accompanying samples submitted for preproduction and periodic production testing), or by applicable documents referenced in Section 2, unless specified in the contract or order (see 6.2).

3.2.2 Preproduction sample. Unless otherwise specified in the contract or order, preproduction samples of the contractors shall be manufactured using the methods and procedures proposed for the production lot. These units shall be tested as specified in Section 4 herein and are for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing contactors that comply with the technical requirements of the contract. See 4.3.1.

3.2.3 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the contactors shall conform strictly to the detailed requirements indicated on the drawings and specifications listed on NAVORD Drawings 877137 and 877138.

3.2.4 Finishes, coatings, construction, and dimensions, Finishes, coatings, construction, and dimensions for the contractors shall be strictly In accordance with the applicable drawings, specifications, and publications listed on NAVORD SD 284040 and LD 284041.

3.2.5 Fungus and corrosion. The contractors shall be fungus and corrosion resistant in accordance with NAVORD OS 6341.

3.2.6 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed on NAVORD LD 284040 and LD 284041.

3.2.7 Threads. Unless otherwise specified, all threads shall-be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

3.2.8 Conflicting requirements. Conflicting requirements arising between this specification or any specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

3.3 Definitions. Terms applicable to this specification are defined in MIL-R-2033.

3.4 Performance requirements and product characteristics. The contectors shall meet the following performance requirements and product

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:16Z
Check the source to verify that this is the current version before use.

characteristics. Unless otherwise specified, all parts and subassemblies herein referred to are listed on NAVORD Drawings 877137 and 877138.

3.4.1 Characteristics common to both the series contactor (NAVORD Drawing 877137) and the paralleling contactor (NAVORD Drawing 877138):

3.4.1.1 Pick-up. The contractors shall pick-up when a maximum of 10 volts dc is applied to the coil (NAVORD Drawing 815178).

3.4.1.2 Seal-in. The contractors shall seal-in so that the armature (NAVORD Drawing 1154934 for the series contactor, or 865968 for the paralleling contactor) is tight against the pole piece (NAVORD Drawing 865924) at an applied voltage between 12 and 17 volts dc.

3.4.1.3 mop-out. The contractors shall drop-out against their stops at a voltage not less than 2.25 volts dc.

3.4.1.4 Rated coil voltage. With 24 ±1 volts dc applied to the coil terminals, the contractors shall close with a fast snappy action, free from sticking, binding, or slow moving armature assembly.

3.4.1.5 Coil (NAVORD Drawing 815178)

3.4.1.5.1 The coil shall have 1400 ±14 turns of number 24 enameled aluminum wire (two conductors in parallel), and the coil shall be free from short-circuited turns.

3.4.1.5.2 The coil shall have a dc resistance of 13 ±1 ohms at 25 degrees centigrade (°C) (77 degrees Fahrenheit (°F)).

3.4.1.5.3 The coil shall be vacuum impregnated and coated with varnish MIL-I-24092, Class 105, Type M. The outside of the coil, and other insulated parts, shall be treated with one coat of fungicidal varnish conforming to MIL-V-173, as set forth on NAVORD Drawing 815178.

3.4.1.5.4 When mounted in the contactor, the temperature rise of the coil shall not exceed 55°C in 30 ±3 minutes of operation, with 24 ±1 volt dc applied to the coil terminals.

3.4.1.6 Contacts

3.4.1.6.1 The contact hinge assembly (NAVORD Drawing 865918) shall be free from binding or drag.

3.4.1.6.2 Upon application of voltage to the coil., or hand pressure to the armature, the contacts shall close so that:

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:16Z
Check the source to verify that this is the current version before use.

3.4.1.6.2.1 When one pair of contacts just closes, the other pair shall meet, or be open not more than 1/32 inch.

3.4.1.6.2.2 When the contacts first meet, the line of contact shall be within 9/32 inch of the toe end of the contact.

3.4.1.6.2.3 As the armature seals in, the moving contacts shall rock.

3.4.1.6.2.4 When the armature is tight against the pole face, the line of contact shall be 1/4 ±1/8 inch from the heel end of each pair of contacts.

3.4.1.7 Insulation

3.4.1.7.1 The insulation used in the contractors shall be of such class that the flow of heat will not destroy any portion of the contactor within the operating life as set forth in 3.4.5.

3.4.1.7.2 Insulating materials falling under the commercial designation of "Silicones" are not acceptable due to the deleterious effect of silicone vapors on commutating machinery.

3.4.1.7.3 Insulation resistance. At the time of manufacture, the contactor shall have a minimum insulation resistance of 100 megohms, at ambient temperature, and a minimum of one megohm resistance at 100°C ±10°C (212°F ±18°F), measured between open contacts and between closed contacts and the contactor frame (ground), and between the coil and the frame.

3.4.1.7.4 Overpotential. The contractors shall withstand, without breakdown, an alternating current (ac) voltage (hi-pot test) of 1150 ±50 volts root mean square (rms) minimum, at 60 Hertz (Hz) applied for 60 ±6 seconds between all current carrying parts (including the coil) and the frame of the contactor.

3.4.2 Series contactor function. When mounted in the operating position with the base vertical and the contacts uppermost:

3.4.2.1 The contactor shall be capable of connecting a battery having an open circuit voltage of 110 volts nominal directly across a series motor so that the peak starting current is 2000 ±200 amperes (amps), falling approximately exponentially to 525 ±25 amps in 1.6 ±0.8 seconds.

3.4.2.2 After starting the motor, the contactor shall carry a current of 525 ±25 amps (at 80 volts nominal) for at least 20 minutes without exceeding a temperature rise of 125°C (257°F) in the current carrying parts.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:16Z
Check the source to verify that this is the current version before use.

3.4.2.3 With the arc shield in place, the contactor shall be capable of interrupting the motor current of 525 ±25 amps at 80 volts minimum. All arcing shall occur at the toe of the contacts so that repeated cycles of the contactor will not be adversely affected.

3.4.3 Paralleling contactor function. When mounted in the operating position with the base vertical and the contacts uppermost:

3.4.3.1 Each section of the contactor shall be capable of connecting, a battery having an open circuit voltage of 55 volts nominal directly across a series motor so-that the peak starting current is 800 ±200 amps, falling approximately exponentially to 165 ±16 amps in 3 ±1.5 seconds.

3.4.3.2 After starting the motor, each section of the contactor shall carry a current of 165 ±15 amps (at 40 volts nominal) for at least 60 minutes without exceeding a temperature rise of 125°C (257°F) in the current carrrying parts.

3.4.3.3 With the arc shield in place, the contactor shall be capable. of interrupting the motor current of 165 ±15 amps at 40 volts minimum. All arcing shall occur at the toe of the contacts so that repeated cycles of the contactor will not be adversely affected.

3.4.4 Duty cycle. A duty cycle shall consist of the operations set, forth in 3.4.2 for the series contactor and 3.4.3 for the paralleling contactor in the sequence given.

3.4.5 Operating life. The operating life of both the series and paralleling contractors shall consist of at least 200 duty cycles as set forth in 3.4.4.

3.5 Environmental requirements. The contractors shall meet the other applicable requirements of this specification during or after the tests defined or specified in 4.5.

3.5.1 Vibration. The contractors shall meet the following vibration requirements:

3.5.1.1 Vibration, nonoperating. The contractors shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.1.2.

3.5.1.2 Vibration, operating. The contractors shall show no damage and shall meet all the performance requirements and product characteristics of this specification during and after the test specified in 4.5.1.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:16Z
Check the source to verify that this is the current version before use.

3.5.1.3 Vibration, transportation. The contractors shall show no damage and shall meet all the performance requirements and product characteristics. of this specification after being tested as in 4.5.1.4.

3.5.2 Shock, transportation. The contractors shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.2.

3.5.3 Temperature and humidity.

3.5.3.1 High temperature and humidity, nonoperating. The contractors shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.5.3.1.

3.5.3.2 High temperature, operating. The contractors shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.5.3.2.

3.5.3.3 Low temperature nonoperating. The contractors shall met all the performance requirements and product characteristics of this specification after being tested as in 4.5.3.3.

3.5.3.4 Low temperature, operating. The contractors shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.5.3.4.

3.6 Screw assemblies. Assembly screws shall be tight. The ward “tight” means that screws shall be tightened in conformance with established standard engineering practices.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:16Z
Check the source to verify that this is the current version before use.

3.7 Riveting. Riveting operations shall be carefully performed in order to ensure that the rivets are tight and satisfactorily headed.

3.8 Cleanup. After final assembly all contractors shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as rosin flash shall be removed.

3.9 Workmanship. The equipment, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner which will ensure its meeting all the requirements set forth herein. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, riveting, machine screw assemblage, welding, brazing, and freedom of parts from burrs and sharp edges.

The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performante of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The . Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the contactors shall be classified as follows:

(a) Reproduction inspection (paragraph 4.5)

(b) Quality conformance inspection (paragraph 4.4)

(c) Periodic production inspection (paragraph 4.5)

4.3 Sampling.

4.3.1 Preproduction. A preproduction sample of five series and five paralleling contractors, manufactured in accordance with 3.2.2, shall be subjected to the preproduction tests detailed in 4.5 at an activity designated by the procuring activity. Acceptance of the preproduction sample

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:16Z
Check the source to verify that this is the current version before use.

shall be based on no defects in the sample. Further production of the contractors by the supplier, prior to the approval of the preproduction sample, shall be at the suppliers risk.

4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate "inspection lot" and my differ from the quantity specified in the total order or contract. In any event, all units of a single "inspection lot" shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.

4.3.3 Periodic production samples. From each succeeding lot or grand lot of 300 or less that have passed the quality conformance inspection of 4.4, a random sample of five series and five paralleling contactors will be selected for submission to an activity designated by the procuring activity to determine compliance of the samples with the requirements of the contract, specification, and drawings. The contractors shall be considered satisfactory if they meet all specification and pertinent drawings requirements and withstand the tests specified in 4.4 and 4.5. Failure of the periodic production sample to comply with these requirements will result in the rejection of the lot, as determined by the procuring activity.

4.4 Quality conformance inspection. Each contactor delivered under contract shall be subjected to the following tests and examinations. Failure. to meet any requirement specified herein shall be considered cause for rejection.

4.4.1 Test conditions. Unless otherwise specified, the contractors shall be subjected to quality conformance tests under the following conditions:

4.4.1.1 Temperature: Room ambient 65°Fto95°F (18°C to 35°C).

4.4.1.2 Altitude: Normal ground.

4.4.1.3 Vibration: None. .

4.4.1.4 Humidity: Room ambient to 95% relative, maximum.

4.4.2 Test and inspection equipment and facilities- The manufacturer shall furnish and maintain all necessary test equipment and facilities "and shall provide personnel for performing all quality conformance tests. The test equipment shall be adequate in quantity and, unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required quality conformance tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:16Z
Check the source to verify that this is the current version before use.

4.4.3. Test equipment. The following items of test equipment are required to perform the quality conformance tests set forth in this specification:

4.4.3.1 One dc voltage source continuously variable from 0 to 30 volts dc at 5 amps maximum.

4.4.3.2 One voltmeter 0 to 30 volts dc, 1% accuracy.

4.4.3.3 One centigrade thermometer encompassing a range of 0 to 175°C with an accuracy of ±1% calibrated in increments not larger than 5°C.

4.4.3.4 One centigrade thermometer encompassing a range of 0 to 60°C with an accuracy of ±1%, calibrated in increments not larger than 1°C.

4.4.3.5 One dc ammeter, 0 to 5 amps, l% accuracy.

4.4.3.6 one ohmmeter encompassing a mange of 5 to 30 ohms, with an accuracy of 0.5% calibrated in increments not larger than 0.1 ohm.

4.4.3.7 One feeler gage, 0.0312 ±0.0001 inch.

4.4.3.8 One 6 inch metal machinist scale, calibrated in 1/32 inch increments.

4.4.3.9 one short-circuited turns tester, including a sensing element to detect and indicate the presence of any short-circuited turns in coil (Dwg 815178).

4.4.3.10 One coil turns counter capable of measuring the number of turns in the coil (Dwg 815178) to within two turns.

NOTE

The instruments specified in 4.4.3.9 and 4.4.3.10 can be packaged as one unit.

4.4.3.11 One megohmmeter capable of measuring insulation resistance at 500 volts dc and including the range of 0-1000 megohms with an accuracy of 2%..

4.4.3.12 One high potential test set capable of producing an ac voltage of 60 Hz continuously variable from 0 to 1500 volts, and containing a sensing element to indicate insulation failure, and a meter to indicate terminal voltage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:16Z
Check the source to verify that this is the current version before use.

4.4.4 Test procedure

4.4.4.1 To test for pick-up, seal-in, and drop-out, as set forth in 3.4.1:

4.4.4.1.1 Connect the power source (4.4.3.1) and voltmeter (4.4.3.2) directly across the terminals of the contactor coil, after mounting the contactor with its base lying in a vertical plane and the contacts uppermost.

4.4.4.1.2 Gradually increase the voltage on the contactor coil and ascertain the voltage at which the armature of the contactor picks-up so that the contacts just close. The pick-up voltage shall be within the limits set forth in 3.4.1.1.

4.4.4.1.3 Gradually increase the voltage on the contactor coil and ascertain the voltage at which the armature pulls up tightly against the pole piece. The seal-in voltage shall be within the limits set forth in 3.4.1.2.

4.4.4.1.4 Gradually reduce the voltage on the contactor coil and ascertain the voltage at which the armature drops out against the stop. The drop-out voltage shall be within the limits set forth in 3.4.1.3.

4.4.4.2 To test for rated coil voltage as set forth in 3.4.l.4:

4.4.4.2.1 Adjust the power source (4.4.3.1) to an output voltage of 24 ±1 volt dc.

4.4.4.2.2 Close the switch, which has been interposed between the power source and the contactor coil, and observe that the action of the contactor conforms to the requirements of 3.4.1.4.

4.4.4.3 To test for number of coil turns and short circuited turns as set forth in 3.4.1.5.1:

4.4.4.3.1 Prior to the assembly of the coil into the contactor, test to assure that the correct number of turns are present in the coil and that there are no short circuited turns as specified in 3.4.1.5.1, using the coil test equipment (4.4.3.9 and 4.4.3.10).

4.4.4.4 To test for dc resistance of the coil as set forth in 3.4.1.5.2:

4.4.4.4.1 Using the ohmmeter (4.4.3.6) measure the dc resistance of the coil and correct the reading to 25°C (77°F). The resistance shall be within the limits specified in 3.4.1.5.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:16Z
Check the source to verify that this is the current version before use.

4.4.4.5 To test for binding or drag inthe contact hinges as set forth in 3.4.1.6.1:

4.4.4.5.1 Manually operate the contact hinges to assure freedom from binding or drag.

4.4.4.6 To test for contact alignment as set forth in 3.4.1.6.2:

4.4.4.6.1 Either by hand pressure, or by application of voltage to the contactor coil, slowly close the contactor and measure the gap in the second pair of contacts, using the feeler gage (4.4.3.7), when the first pair of contacts just meet. The gap shall not exceed that specified in 3.4.1.6.2.1.

4.4.4.6.2 (Continuing the pressure set forth in 4.4.4.6.1, measure the line of initial contact using the machinists scale (4.4.3.8). The position of the initial line of contact shall conform to that specified in 3.4.1.6.2.2.

4.4.4.6.3 Increase the pressure on the contactor armature and observe that as the armature seals in, the line of contact shall shift toward the heel of the contacts and that the contacts rock as set forth in 3.4.1.6.2.3.

4.4.4.6.4 With the armature held tightly against the pole face either by hand pressure or electrically, measure the position of the line of contact, in each pair of contacts, by use of the machinists scale (4.4.3.8) by sighting through the gap between the contact. The location of the line of contact shall be as set forth in 3.4.1.6.2.4.

4.4.4.7 To test for insulation resistance as set forth in 3.4.1.7.3:

4.4.4.7.1 Using the megohmmeter (4.4.3.11) measure the insulation resistance at ambient temperature between the following points:

4.4.4.7.1.1 With the ground (earth) terminal connected to the contactor frame and the line terminal connected to the coil terminals.

4.4.4.7.1.2 With the ground (earth) terminal connected to the contactor frame and line terminal connected to all contacts (both moving and stationary).

4.4.4.7.1.3 With the ground (earth) terminal connected to one stationary contact and the line terminal. connected to the other stationary contact.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:16Z
Check the source to verify that this is the current version before use.

4.4.4.7.1.4 With the ground (earth) terminal connected to both stationary contacts and the line terminal connected to both moving contacts.

4.4.4.7.1.5 On the paralleling contactor only, with the ground (earth) terminal connected to one moving contact and the line terminal connected to the other moving contact.

4.4.4.7.2 The insulation resistance as measured shall be within the limits specified-for insulation resistance at ambient temperature in 3.4.1.7.3.

4.4.4.8 To test for overpotential as set forth in 3.4.1.7.4:

4.4.4.8.1 Connect one lead of the high potential test set (4.4.3.32) to the contactor frame and the other lead simultaneously to all of the contacts and one of the coil terminals.

4.4.4.8.2 Gradually increase the voltage from O to 1150 ±50 volts at a rate not exceeding 150 volts per second, and hold at the attained voltage for 60 ±6 seconds.

4.4.4.8.3 Carefully observe for any evidence of insulation failure to assure conformance with 3.4.1.7.4.

4.4.4.9 Visual and dimensional examinations. Each contactor shall be examined to determine conformance with the requirements for workmanship and dimensions.

4.4.4.10 Packaging, packing, and marking. Packaging, packing, and marking shall comply with the requirements of Section 5.

4.5 Preproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.4, shall be subjected to the following tests and examinations. Certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the procuring activity.

4.5.1 Vibration

4.5.1.1 Mounting. The contactor, less pack or packaging, shall be mounted to a vibration device by means of its mounting holes and vibration tested as specified in 4.5.1.2 and 4.5.1.3.

4.5.1.2 Vibration, nonoperating. The contractors shall be subjected to the “Vibration, nonoperating” test of MIL-E-82590 and thereafter shall meet the requirements of 3.5.1.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:16Z
Check the source to verify that this is the current version before use.

4.5.1.3 Vibration, operatIing. The contractors shall be subjeccted to the "Vibration, operating" test of MIL-E-82590(OS) and during and thereafter shall meet the requirements of 3.5.1.2.

4.5.1.4 Vibration, transportation. The contractors shall be subjected to the "Vibration, transportation" test of MIL-E-82590(OS) and thereafter shall meet the requirements of 3.5.1.3.

4.5.2 Shock, transportation. The contactors shall be subjected to the "Shock, transportation" test of MIL-E-82590(OS) and thereafter shall meet the requirements of 3.5.2.

4.5.3 Temperature and humidity

4.5.3.1 High temperature and humidity, nonoperating. The contactors shall be subjected to the "High temperature and humidity, nonoperating". test of MIL-E-82590(OS) and thereafter shall met the requirements of 3.5.3.1.

4.5.3.2 High temperature, operating. The contractors shall be subjected to the "High temperature, operating" test of MIL-E-82590(OS) and, while at the test temperature, shall meet the requirements of 3.5.3.2.

4.5.3.3 Low temperature, nonoperating. The contractors shall be subjected to the "Low temperature, nonoperating" test of MIL-E-82590(OS) and thereafter shall meet the requirements of 3.5.3.3.

4.5.3.4 Low temperature, operating. The contractors shall be subjected to the "Low temperature, operating" test of MIL-E-82590(OS) and, while at the test temperature, shall meet the requirements of 3.5.3.4.

4.5.4 Duty cycle

4.5.4.1 Mount the contactors in Battery Compartment 877132 or on a frame simulating mounting in a battery compartment with bus work and arc shields in place.

4.5.4.2 Supply power from Silver-Zinc-Alkali Battery MIL-B-18855 or from any other battery or motor generators having the same voltage, internal resistance, and load characteristics.

Source: https://assist.dla.mil -- Downloaded: 202610-18T12:16Z
Check the source to verify that this is the current version before use.

NOTE

External resistance maybe added to lead acid storage batteries or generators to simulate internal resistance and load characteristics of the Silver-Zinc-Alkali Battery MIL-B-18855.

4.5.4.3 Electrical load shall be one of the following:

4.5.4.3.1 Motor MIL-M-18848, mounted on its associated load test stand (section 4.4.2 of MIL-M-18848).

4.5.4.3.2 Any series dc motor hating the same electrical characteristics as the motor specified in MIL-M-18848.

4.5.4.3.3 A combination of resistances, reactance, and fuses to simulate the starting, running, and interruption characteristics of the motor current.

4.5.4.4 Operate each contactor through the cycles specified in 3.4.4, allowing the equipment to cool to within 5°C (9°F) of ambient between cycles.

4.5.4.5 Measure the temperature in the current carrying parts during each cycle. It shall not exceed that set forth in 3.4.4.

4.5.5 Operating life

4.5.5.1 Operate each contactor for 200 duty cycles, as specified in 4.5.4, without malfunction.

4.5.5.2 Successful completion of this test indicates that the requirements for operating life as specified in 3.4.5 have been met.

4.5.6 Temperature

4.5.6.1 With the contactor coil mounted in the contactor, and at ambient temperature, apply 24 ±1 volt dc to the contactor coil terminals for 30 ±3 minutes.

4.5.6.2 The temperature rise as measured by the thermometer (4.4.3.3) shall not exceed 55°C (99°F), as set forth in 3.4.1.5.4.

4.5.7 Insulation resistance

4.5.7.1 Place contractors to be tested in a circulating air oven, operating at 100 ±10°C, for one hour minimum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:16Z
Check the source to verify that this is the current version before use.

4.5.7.2 Measure the insulation resistance of the contactors, either while in the oven or within 10 minutes of their having been removed from the oven, between the points set forth in 4.4.4.7.1.

4.5.7.3 The insulation resistance as measured shall be within the limits specified in 3.4.1.7.3, for insulation resistance at 100°C.

4.6 Waiver of tests. Tests specified in 4.5.1.4 and 4.5.2 above may be waived by the procuring agency for Level C packs only, provided the contractor guarantees safe delivery to the using activity.

4.7 FAunctional tests. At the conclusion of the tests specified in 4.5.1.4 and 4.5.2 above, functional and other tests, as applicable, shall be conducted on the contactors to determine freedom from operational mal-function and compliance with all perfornmce requirements and product characteristics of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The levels of preservation and parking aging shall be as specified (see 6.2).

501.1 Level A

5.1.1.1 unit packaging. The contactor shall be packaged in accordance, with Method IIB of MIL-P-U6.

5.1.2 Level Be Not applicable.

5.1.3 Level C

5.1.3.1 unit packaging The contactor shall be packaged in accordance , with Method III of MIL-P-116.

5.2 Packing. Packing shall be Level A or Level C, as specified (see 6.2).

5.2.1 Level A

5.2.1.1 Consolidation container. The contractor$_s$) packaged as described in 5.1.1.1, shall be packed in a wood box in accordance with PPP-B-621, Class 2.

5.2.1.2 Palletizing$_e$ When specified, Level A packs shall be palletized in accordance with MIL-STD-147, Pattern No. 94.

5.2.2 Level B. Not applicable.

Source: https://assist.dla.mil -- Downloaded: 2026l0-18T12:16Z
Check the source to verify that this is the currentversion before use.

5.2.3 Level C. The contractors, packaged in accordance with 5.1.3.1, shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Level A or C cushioning. The unit package shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices to insure the contractors will meet all the requirements of Section 3 after being tested in accordance with 4.5.1.4 (Vibration) and 4.5.2 (Shock). At the time of packaging or packing, the moisture content of the cushioning material shall be not more than 5 percent.

5.4 Marking.

5.4.1 Special markings. None, unless otherwise specified.

5.4.2 Normal markings. In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, including serial numbers.

6. NOTES

6.1. Intended use. The contractors are intended to be used in Torpedo Mk 37 Mods 0, 2, and 3.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) ‘Any exceptions to this specification.

(c) Required level of preservation and packaging (see 5.1).

(d) Required level of packing, including palletization (see 5.2).

(e) Assigned activity for preproduction and periodic production inspections.

(f) Data required other than that specified in 3.2.1.

(g) That additional samples required because of failure of any unit(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:16Z
Check the source to verify that this is the current version before use.

6.2.1 The attention of the contracting officer is directed to 4.6 of this specification.

6.3 Preparation for delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as follows:

Level A. This level shall be used for those items which are to be shipped indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

Level C. This level shall be used only when it is definitely known that the packaged items are to be shipped to domestic installations for immediate use at the first receiving activity.

Preparing Activity
Navy - OS
Project No. 1355-N197.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:16Z
Check the source to verify that this is the current version before use.